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A unified model for species fluxes, ion impact energy and heat transfer at the electrode of an atmospheric-pressure arc discharge

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Abstract

1 Introduction

Arc discharges find extensive applications in welding, cutting, and thermal spraying. In recent years, their use has expanded to emerging applications, including plasma pyrolysis, advanced material synthesis, and metallurgical reduction processes [1, 2, 3]. An atmospheric-pressure arc discharge spans a wide range of length scales, differing by several orders of magnitude. The millimeter-to-centimeter-scale quasi-neutral arc column constitutes the bulk of the discharge, where convection, thermal conduction, Joule heating, and radiation interact under conditions that are often close to local thermodynamic equilibrium (LTE). In contrast, the sub-micron to sub-millimeter regions adjacent to the electrodes are characterized by strong nonequilibrium effects. In these regions, the plasma is subjected to the relatively cold electrode surfaces, steep density gradients associated with particle absorption at the electrodes, and strong electric fields arising from charge separation. These fields can accelerate electrons and ions to energies substantially higher than their local thermal energies.

Resolving the full range of length scales throughout the entire discharge is neither computationally practical nor necessary for many applications. Consequently, arc-discharge models commonly treat the quasi-neutral bulk plasma separately from the nonequilibrium electrode regions. In the bulk, the quasineutrality condition, $n_i \approx n_e$, eliminates the need to explicitly solve Poisson's equation, while the near-LTE assumption further reduces model complexity by allowing thermodynamic and transport properties to be determined primarily from the local temperature and pressure. Macroscopic effects such as cathode jets, electrode geometry, convection, and electromagnetic forces can then be incorporated within a continuum formulation based on the Navier–Stokes equations with Joule-heating and Lorentz-force terms. The non-equilibrium electrode regions are generally modeled to account for diffusion, reaction, sheath effects in the discharge.

Quasi-neutral bulk plasma modeling, can be broadly classified into four categories according to the degree of thermodynamic and chemical equilibrium assumed and the treatment of electron and heavy-particle temperatures.

****(i) Full local thermodynamic equilibrium.**** Experimental measurements of species temperatures and electron densities indicate that the arc core is generally close to LTE [4, 5, 6]. In a full LTE model, electrons and heavy particles are characterized by a single temperature T, with Maxwellian velocity distributions, Boltzmann excitation, Saha ionization, and Dalton's law used to determine the plasma state and properties. The governing equations typically consist of mass, momentum, energy, and current continuity [7, 8, 9]. Conventional LTE models does not account for electrode falls and sheaths effects.

****(ii) Two-temperature models.**** Two-temperature (2T) models treat electrons and heavy particles as separate Maxwellian populations at T_e and T_h, coupled through elastic energy exchange. In two-temperature-equilibrium (2T-EQ) models, the chemical composition is additionally determined from a generalized Saha relation for $T_e \neq T_h$. Hsu and Pfender [10] showed that $T_e \approx T_h$ in the high-density arc core, with significant temperature separation emerging in the low-density fringes and near the anode. However, the generalized Saha closure is not unique: different formulations can predict electron densities differing by up to an order of magnitude away

from $T_e/T_h \approx 1$ [11]. Furthermore, inconsistencies in the partitioning of ionization-energy and reactive thermal-conductivity terms between the electron and heavy-particle energy equations have been identified by Freeton et.al [12] in several formulations.

****(iii) Nonequilibrium models.**** Fully nonequilibrium models abandon equilibrium closures such as LTE and generalized Saha equilibrium, allowing the plasma composition and electron and heavy-particle temperatures to evolve independently. These models typically include multicomponent species transport and separate electron and heavy-particle energy equations. Examples include the nonequilibrium arc models of Haidar [13], Gonzalez *et al.* [14], and Amakawa *et al.* [15]. By resolving departures from equilibrium directly, these models can describe the transition from the LTE-like bulk plasma toward the strongly nonequilibrium electrode region. However, the space-charge sheath is generally still treated separately using an appropriate sheath model.

****(iv) LTE with diffusion-driven ionization nonequilibrium.**** Another approach considered in the literature retains $T_e = T_h = T$ while allowing the electron and ion densities to depart from their local Saha-equilibrium values [16, 17, 18]. This departure arises when diffusion competes with the ionization–recombination relaxation processes, particularly near the electrodes. Lowke [16] demonstrated this approach using a full-diffusion calculation that solves the ambipolar electron-continuity equation together with current continuity, including the electron-diffusion contribution. The resulting solution produces a retarding-field potential structure near the electrode that is absent in a conventional LTE calculation. However, because $T_e = T_h$ is retained throughout the domain, this formulation cannot capture the strong electron–heavy-particle temperature separation expected in the low-density electrode-adjacent region.

1.1 Coupling the bulk plasma to the electrode: sheath models

Regardless of the bulk-plasma model employed, the bulk solution must ultimately be coupled to a description of the electrode region. The appropriate coupling strategy depends on the assumptions underlying the bulk formulation and, in particular, on the degree of nonequilibrium that is retained. For fully LTE models (i), the electrode region must account for the breakdown of LTE, including the separation between electron and heavy-particle temperatures, deviations from Saha equilibrium, and the formation of a space-charge sheath. For two-temperature and two-temperature-equilibrium models (ii), the bulk formulation already accounts for electron–heavy-particle temperature nonequilibrium; however, deviations from ionization equilibrium and the formation of a space-charge sheath must still be described. In fully nonequilibrium bulk models (iii), an additional ionization layer or presheath is not necessarily required, since departures from equilibrium are already permitted within the bulk formulation. The principal remaining transition to the electrode is therefore associated with the breakdown of quasineutrality and the formation of the space-charge sheath.

A common approach is to construct the electrode-region model from separate submodels for the ionization layer or presheath and the sheath. The ionization layer is treated as a collisional region and coupled to the sheath, which is treated as collisionless using an analytical model [19, 20, 21, 22]. This approach avoids the direct solution of the stiff Poisson equation across the sheath. However, Benilov *et al.* [23] pointed out that such matching is strictly valid only if the sheath is collision-dominated.

Almeida, Benilov, and Naidis [24] evaluated the number of ion collisions occurring as an ion traverses the cathode space-charge sheath,they pointed out that for atmospheric-pressure argon arc discharge, the ion-collision number is far greater than unity rather than $\ll 1$. Which demonstrate that a physically consistent electrode model for high-pressure arc discharges should account for collisional ion transport within the sheath. A kinetic study performed by Li Sun et al. [25] for the near-cathode region supported the collision dominated view. Futher Semenov et al. [26] demonstarted that the simiple sheath models that assume a bohm velocity at the sheath edge are not valid for anode-region of high-pressure arc discharges.

An alternative approach is *unified modeling*, in which a single set of governing equations, including Poisson's equation, is solved across the entire interelectrode region without explicitly separating the bulk plasma from the sheath. This approach provides a self-consistent description of the electrode region but is computationally expensive because the mesh must resolve the extremely small sheath length scales. Consequently, applications to arc discharges have largely been restricted to one-dimensional calculations extending from the electrode into the bulk plasma [24, 27, 26].

Except in the immediate vicinity of the electrodes, an atmospheric-pressure arc is sufficiently close to local thermodynamic equilibrium (LTE) that the bulk plasma can be modeled using a single temperature. Consequently, nonequilibrium effects need not be considered in the bulk, which

substantially reduces the complexity of the model.The approach adopted in the present work therefore retains the computationally efficient LTE treatment for the bulk of the arc while explicitly resolving the nonequilibrium physics in the near-electrode region. Rather than prescribing electrode-boundary quantities or relying on an analytical sheath model, a unified drift-diffusion–Poisson formulation is employed to simultaneously determine the charged-particle distributions, species fluxes, ion impact energies, heat transfer to the electrode, and sheath voltage drop. The same governing drift-diffusion–Poisson system is solved continuously from the quasi-neutral bulk plasma through the near-electrode transition region and into the sheath. The highly disparate spatial scales are resolved using adaptive mesh refinement (AMR), with the computational mesh refined to a fraction of the local Debye length in the sheath region.

An additional advantage is that one need not worry about field reversals. Field reversals and changes in the sign of the potential drop near the anode have been reported in atmospheric-pressure arcs and can depend on several factors, including the attachment mode, current density, and electrode ablation effects [28, 29, 30]. By solving the drift-diffusion–Poisson system self-consistently across the near-electrode region, the sheath potential structure and its polarity emerge from the coupled solution rather than being assumed a priori. This approach enables one to use the results from the computational efficient bulk plasma treatment to inform the boundary conditions for the near-electrode region, while simultaneously capturing the nonequilibrium effects that arise in the sheath and ionization layer. The model and its numerical implementation in the PlasmeX code are described in Section 4.

2 Physical model and scale analysis

2.1 Governing equations

In the electrode region, the plasma is in contact with a relatively cold, absorbing electrode surface. This interaction produces strong gradients in the plasma properties, and the plasma departs progressively from LTE as the electrode is approached. We estimate the length scales over which these departures develop by considering the LTE plasma adjacent to the electrode region at temperatures between $T = 12{,}000$ K and $T = 22{,}000$ K, which are typical of the anode and cathode regions, respectively, of a high-intensity atmospheric-pressure arc discharge [16, 9, 31].

In the bulk plasma, the electron density n_e is governed by ionization and recombination reactions. The characteristic time scale of these reactions can be obtained by linearizing $\dot{n}_e = k_{\text{ion}} n_e n_n - k_{\text{rec}} n_e^3$ (with $n_i = n_e$) about equilibrium at fixed p and T (so that $\delta n_n = -2\,\delta n_e$), which gives $\tau_{rxn} = 1/(2k_{\text{rec}} n_e^2 + 2k_{\text{ion}} n_e)$. A characteristic length scale for the reactions is then $\lambda_{rxn} = \sqrt{D_a \tau_{rxn}}$, where D_a is the ambipolar diffusion coefficient. This length scale implies that a plasma out of chemical equilibrium needs a distance of order λ_{rxn} to relax back to equilibrium. Similarly, since an electron loses only a fraction $2m_e/M$ of its energy in an elastic collision with a heavy particle, the length scale for the energy relaxation between electrons and heavy species can be estimated as $\lambda_T = 1/[\sqrt{2m_e/M}\,(n_i \overline{Q}_{ei} + n_n \overline{Q}_{en})]$, where n_i and n_n are the ion and neutral number densities, and $\overline{Q}_{ei}$ and $\overline{Q}_{en}$ are the averaged momentum-transfer cross sections for electron–ion and electron–neutral collisions. $\overline{Q}_{en}$ is the Maxwellian average of the argon momentum-transfer cross section used in Appendix 3. Electron–ion collisions are Coulomb collisions, for which the averaged momentum-transfer cross section is $\overline{Q}_{ei} = e^4 \ln\Lambda/[32\pi\varepsilon_0^2 (k_B T_e)^2]$ [24]. Here, k_B is the Boltzmann constant and $\ln\Lambda$ is the Coulomb logarithm, $\ln\Lambda = \ln\left(1.24 \times 10^7 T_e^{3/2} n_e^{-1/2}\right)$, with T_e in K and n_e in m^{-3}. Ions and atoms have equal masses and exchange energy efficiently the length scale for the equilibration of the heavy-species temperature is $\lambda_{T_h} = 1/[(n_i + n_n)\overline{Q}_{in}]$. Here $\overline{Q}_{in} = 2[7 \times 10^{-10} - 0.6 \times 10^{-10} \ln(2k_B T_h/e)]^2$ m^2 is the ion–neutral momentum-transfer cross section [32].

At the smallest scale, the Debye length calculated from the LTE number densities, $\lambda_{De} = \sqrt{\varepsilon_0 k_B T_e/(n_e e^2)}$, is of order 10^{-8} m. It sets the extent over which charge separation and space-charge effects are important. These scales are reported in Table 1, together with electron mean free path $\lambda_e = 1/(n_n \overline{Q}_{en} + n_i \overline{Q}_{ei})$ and ion mean free path $\lambda_i = 1/(n_n \overline{Q}_{in} + n_i \overline{Q}_{ii})$, where $\overline{Q}_{ii}$.

The resulting hierarchy of length scales is $\lambda_{De} \ll \lambda_i \sim \lambda_e \sim \lambda_{T_h} < \lambda_{\text{rxn}} \sim \lambda_T$. These scale estimates indicate that a continuum description remains valid to a first approximation in the quasi-neutral plasma and is reasonably applicable within the sheath. Furthermore, the gas temperature near the anode varies over a fluid boundary-layer thickness of order $\sim$ mm, which is much larger than λ_{T_h}. Thus, the heavy species, consisting of ions and neutrals, can reasonably be described using a single temperature. These scales also indicate that the effect of electrodes is felt over a region of thickness of order 10^{-4} m by plasma, in agreement with other modelling results and analyses [24, 19, 22, 20]. Consistent with this estimate, Lowke and Tanaka [16] showed that excluding a near-electrode region of this thickness is sufficient for the remaining plasma to be

Table 1. Characteristic scales of LTE argon at atmospheric pressure ($T_e = T_h = T$, Saha equilibrium), ordered from left to right by increasing size ($\lambda_i \approx \lambda_e$, and $\lambda_{rxn} \approx \lambda_T$ at 12 000 and 14 000 K). The reaction scales use the rate coefficients of Hoffert and Lien [33].

T (K)	λ_{De} (m)	λ_i (m)	λ_e (m)	λ_{T_h} (m)	λ_{rxn} (m)	λ_T (m)	τ_{rxn} (s)
12 000	3.0×10^{-8}	8.1×10^{-7}	1.2×10^{-6}	2.1×10^{-6}	2.3×10^{-4}	2.3×10^{-4}	1.2×10^{-5}
14 000	2.1×10^{-8}	6.6×10^{-7}	7.5×10^{-7}	3.2×10^{-6}	1.5×10^{-4}	1.4×10^{-4}	3.0×10^{-6}
18 000	2.1×10^{-8}	8.8×10^{-7}	8.9×10^{-7}	5.9×10^{-6}	8.8×10^{-5}	1.7×10^{-4}	5.1×10^{-7}
20 000	2.3×10^{-8}	1.1×10^{-6}	1.1×10^{-6}	6.9×10^{-6}	6.9×10^{-5}	2.2×10^{-4}	2.5×10^{-7}
22 000	2.5×10^{-8}	1.4×10^{-6}	1.4×10^{-6}	7.8×10^{-6}	5.6×10^{-5}	2.8×10^{-4}	1.4×10^{-7}

modelled as being in LTE. This separation of scales necessitates treating the near-electrode region separately from the bulk plasma.

We therefore model the electrode region as a one-dimensional domain of length 5×10^{-4} m, with the electrode surface at one end and the bulk plasma at the other. Externally imposed bulk flows, such as the cathode jet, are not represented. The heavy gas is nevertheless not assumed to be at rest, since it can move during the transient stages of the discharge. The plasma is taken to consist of three species: electrons (e), singly charged argon ions (i), and argon atoms (n). Multiply charged ions are neglected in the anode region because of the prevailing low temperatures which are not sufficient for their ionization, in cathode region however they are not negligible but rather have a negligible effect, as justified in [34]. Each species is assumed to follow a Maxwellian velocity distribution (see Appendix 3).

The solutions are soughted for the species number densities, the electron and heavy-particle temperatures, and the electric field. The folloowing governing equations are thus solved in the electrode region.

The number densities of the three species obey

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \mathbf{\Gamma}_e = S, \qquad \frac{\partial n_i}{\partial t} + \nabla \cdot \mathbf{\Gamma}_i = S, \qquad \frac{\partial n_n}{\partial t} + \nabla \cdot \mathbf{\Gamma}_n = -S, \tag{1}$$

Different authors have evaluated the net ionization rate S in different ways [24, 19, 16, 31]. In this work we use the reaction rates as proposed by Almeida et al. [24], where ionization is limited by the excitation of the first excited level, and accounts for the radiative escape of resonance radiation when $n_e \lesssim 10^{21}$ m^{-3}.

The momentum equation for the plasma as whole is given by $\rho\, D\mathbf{u}/Dt = \mathbf{F}$, where $\mathbf{F}$ is the net force on the plasma. For each species s, the corresponding momentum equation is $m_s n_s\, D_s\mathbf{u}_s/Dt = \mathbf{F}_s$, where $\mathbf{F}_s$ is the net force on the species. We neglect the inertia of each species relative to the heavy gas, $D_s\mathbf{u}_s/Dt \simeq D\mathbf{u}/Dt$, so that every species shares the acceleration of the heavy gas and there is no relative acceleration between species. The inertia of species s is therefore $Y_s\mathbf{F}$, where $Y_s = m_s n_s/\rho$ is its mass fraction. The momentum equations then reduce to the local force balances [35]

$$0 = -en_e\mathbf{E} - \nabla p_e - m_e n_e \bar{\nu}^M_{en}(\mathbf{u}_e - \mathbf{u}_n) - m_e n_e \bar{\nu}^M_{ei}(\mathbf{u}_e - \mathbf{u}_i) - C_e n_e k_B \nabla T_e - Y_e\mathbf{F}, \tag{2}$$

$$0 = +en_i\mathbf{E} - \nabla p_i - Mn_i \bar{\nu}^M_{in}(\mathbf{u}_i - \mathbf{u}_n) - Mn_i \bar{\nu}^M_{ie}(\mathbf{u}_i - \mathbf{u}_e) + C_i n_e k_B \nabla T_e - Y_i\mathbf{F}, \tag{3}$$

$$0 = -\nabla p_n - Mn_n \bar{\nu}^M_{ni}(\mathbf{u}_n - \mathbf{u}_i) - Mn_n \bar{\nu}^M_{ne}(\mathbf{u}_n - \mathbf{u}_e) + C_n n_e k_B \nabla T_e - Y_n\mathbf{F}. \tag{4}$$

The frequencies $\bar{\nu}^M_{en}$, $\bar{\nu}^M_{ei}$, $\bar{\nu}^M_{in}$, $\bar{\nu}^M_{ie}$, $\bar{\nu}^M_{ni}$, and $\bar{\nu}^M_{ne}$ are *averaged momentum-transfer collision frequencies* and are calculated as shown in Appendix 3. C_e is the electron thermal-diffusion coefficient, and C_i and C_n are its reactions on the ions and atoms. Momentum conservation in each collision pair relates the frequencies and the thermal-diffusion forces sum to zero.

$$m_e n_e \bar{\nu}^M_{en} = Mn_n \bar{\nu}^M_{ne}, \qquad m_e n_e \bar{\nu}^M_{ei} = Mn_i \bar{\nu}^M_{ie}, \qquad n_i \bar{\nu}^M_{in} = n_n \bar{\nu}^M_{ni}, \qquad C_i + C_n = C_e \tag{5}$$

Adding equations (2)–(4) and using equation (5) results in the total momentum equation.

$$\rho\, D\mathbf{u}/Dt = e(n_i - n_e)\mathbf{E} - \nabla p. \tag{6}$$

In practice we solve equation (2), (3), and (6) instead of (4). We need a fourth equation to determine all four velocities, ($\mathbf{u}_e$, $\mathbf{u}_i$, $\mathbf{u}_n$, and $\mathbf{u}$). This is provided by the definition of the heavy-gas velocity $\mathbf{u}$ as the mass-weighted average of the ion and neutral velocities,

$$\mathbf{u} = \frac{n_i\mathbf{u}_i + n_n\mathbf{u}_n}{n_i + n_n}. \tag{7}$$

The electron balance (2) is solved for electron flux, with $\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M$ the total electron momentum-transfer frequency and $\nabla p_e = k_B T_e \nabla n_e + n_e k_B \nabla T_e$, gives the electron flux

$$\mathbf{\Gamma}_e = -\frac{e n_e}{m_e \bar{\nu}_e^M}\mathbf{E} - \frac{k_B T_e}{m_e \bar{\nu}_e^M}\nabla n_e - (1 + C_e)\,\frac{n_e k_B}{m_e \bar{\nu}_e^M}\nabla T_e + \frac{n_e}{\bar{\nu}_e^M}\Big(\bar{\nu}_{en}^M \mathbf{u}_n + \bar{\nu}_{ei}^M \mathbf{u}_i\Big), \tag{8}$$

where the negligible term $Y_e\mathbf{F}$ has been dropped. The first two terms are the conventional drift and diffusion. Their coefficients, $e/(m_e\bar{\nu}_e^M)$ and $k_B T_e/(m_e\bar{\nu}_e^M)$, are the magnitudes of the electron mobility and diffusivity; because both are set by the same frequency, they satisfy the Einstein relation identically. The third term combines the temperature-gradient part of the electron pressure force with the thermal-diffusion (Soret) force C_e. The last term is the collisional drag exerted on the electrons by the moving heavy species.

The ion flux follows from the ion balance (3) in the same way. With $\bar{\nu}_i^M = \bar{\nu}_{in}^M + \bar{\nu}_{ie}^M$ the total ion momentum-transfer frequency and $\nabla p_i = k_B T_h \nabla n_i + n_i k_B \nabla T_h$,

$$\mathbf{\Gamma}_i = \frac{e n_i}{M\bar{\nu}_i^M}\mathbf{E} - \frac{k_B T_h}{M\bar{\nu}_i^M}\nabla n_i - \frac{n_i k_B}{M\bar{\nu}_i^M}\nabla T_h + C_i\,\frac{n_e k_B}{M\bar{\nu}_i^M}\nabla T_e - \frac{n_i}{n_h}\frac{\mathbf{F}}{M\bar{\nu}_i^M} + \frac{n_i}{\bar{\nu}_i^M}\Big(\bar{\nu}_{in}^M \mathbf{u}_n + \bar{\nu}_{ie}^M \mathbf{u}_e\Big). \tag{9}$$

The first three terms are ion drift, diffusion, and diffusion driven by the heavy-particle temperature gradient, with mobility $e/(M\bar{\nu}_i^M)$ and diffusivity $k_B T_h/(M\bar{\nu}_i^M)$. The fourth term is the reaction of the ions to the electron thermal-diffusion force. The fifth term is the ions' share, $Y_i\mathbf{F}$, of the net force on the plasma. This share accelerates the ions together with the rest of the heavy gas. The last term is the collisional drag exerted on the ions by the atoms and electrons.

The velocities in the drag terms of equations (8) and (9) are those of the other species, so the two fluxes are coupled. If the drag, thermal-diffusion, and mixture-acceleration terms are removed ($C_e = C_i = 0$, $Y_i\mathbf{F} = 0$, and $\mathbf{u}_s = \mathbf{u} = 0$ in the drag terms), equations (8) and (9) reduce to the classical drift–diffusion fluxes, with mobilities $e/(m_e\bar{\nu}_e^M)$ and $e/(M\bar{\nu}_i^M)$. Unlike the classical fluxes, however, they retain the full pressure-gradient force of each species in a non-isothermal plasma.

The electric field is obtained from Poisson's equation,

$$\nabla\cdot(\varepsilon_0\mathbf{E}) = e(n_i - n_e), \qquad \mathbf{E} = -\nabla\phi, \tag{10}$$

which is solved over the entire computational domain. The transition from the quasi-neutral bulk plasma to the space-charge region is thereby captured without imposing the sheath boundaries a priori.

The electron thermal energy density $\mathcal{E} = \frac{3}{2}n_e k_B T_e$ obeys

$$\frac{\partial \mathcal{E}}{\partial t} + \nabla\cdot\mathbf{h}_e = -e\,\mathbf{\Gamma}_e\cdot\mathbf{E} - \frac{3}{2}n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^E - \varepsilon_{\text{ion}} S, \tag{11}$$

with the electron energy flux

$$\mathbf{h}_e = \frac{5}{2}k_B T_e\,\mathbf{\Gamma}_e - \kappa_e\nabla T_e + k_B T_e\,n_e\Big[C_i(\mathbf{u}_e - \mathbf{u}_i) + C_n(\mathbf{u}_e - \mathbf{u}_n)\Big]. \tag{12}$$

The terms on the right-hand side of equation (11) are Joule heating by the field, elastic energy transfer to the heavy species, and the ionization-energy sink. In the elastic term, $\bar{\nu}_{eh}^E = \bar{\nu}_{en}^E + \bar{\nu}_{ei}^E$, where the superscript E denotes the energy-exchange average of collision frequencies (Appendix 3). The ionization sink is the net ionization rate multiplied by $\varepsilon_{\text{ion}} = 15.76$ eV, the first ionization potential of argon. The energy flux (12) is the sum of the enthalpy flux at $\frac{5}{2}k_B T_e$ per electron, Fourier conduction with the zero-current electron, thermal conductivity κ_e, and a thermal-diffusion heat flux [36].

The common heavy-particle temperature is governed by

$$n_h c_v\left(\frac{\partial T_h}{\partial t} + \mathbf{u}\cdot\nabla T_h\right) = \nabla\cdot(\kappa_h\nabla T_h) + \frac{3}{2}n_e k_B (T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^E + Q_{in} - n_h k_B T_h\,\nabla\cdot\mathbf{u}, \tag{13}$$

where $c_v = \frac{3}{2}k_B$ is the translational heat capacity per heavy particle and κ_h is the heavy-particle thermal conductivity (Appendix 3). The elastic-exchange term is the same as in the electron energy equation (11), but with the opposite sign. The ionization energy is charged entirely to the electrons and does not appear here. The ion–neutral frictional heating is

$$Q_{in} = M n_i\,\bar{\nu}_{in}^M\,|\mathbf{u}_i - \mathbf{u}_n|^2, \tag{14}$$

which is positive definite and reduces to the Joule dissipation of the ion drift when the neutral gas is at rest. It differs from $\mathbf{E}\cdot\mathbf{j}_i$, which can be negative in a strong sheath field, where the field does work against ions diffusing up their own density gradient. The last term in equation (13) is the compression work of the heavy gas.

Finally, the heavy-gas velocity is advanced by an inviscid single-fluid momentum equation,

$$\frac{\partial(\rho\mathbf{u})}{\partial t}+\nabla\cdot(\rho\mathbf{u}\mathbf{u})=-\nabla p+e(n_i-n_e)\mathbf{E},\qquad p=n_ek_BT_e+n_hk_BT_h, \tag{15}$$

that is, $\rho\,D\mathbf{u}/Dt=\mathbf{F}$, with $\mathbf{F}$ the net force (6). This equation governs the motion of the heavy gas as a whole. It also supplies the mixture acceleration whose shares $Y_s\mathbf{F}$ enter the species balances (2)–(4), which in turn govern only the motion of each species relative to the heavy gas. Its conservative form follows from heavy-particle mass conservation: adding the ion and neutral continuity equations (1) and using the flux (**??**) gives $\partial n_h/\partial t+\nabla\cdot(n_h\mathbf{u})=0$, so heavy-particle mass is conserved exactly, since ionization only converts atoms into ions.

The state advanced in time is therefore $\{n_e,n_i,n_n,T_e,T_h,\mathbf{u},\phi\}$. It is governed by the continuity equations (1)–(**??**) with the fluxes (8), (9), and (**??**); Poisson's equation (10); the electron and heavy-particle energy equations (11)–(14); and the heavy-gas momentum equation (15).

2.2 Boundary conditions

The system is closed by conditions at the two ends of the domain: the electrode surface, $x=0$, and the edge of the near-electrode layer, $x=L$, where the solution is matched to the adjacent LTE bulk plasma. Below, $\hat{\mathbf{n}}$ denotes the unit normal at the electrode pointing into the wall. The anode and cathode cases employ the same set of boundary conditions, with the corresponding boundary values specified independently in each case. Thermionic emission is additionally considered at the cathode.

Electrode, $x=0$. The potential is referenced to zero at the electrode, $\phi(0)=0$. Electrons reaching the surface are absorbed at the one-sided (Maxwellian) thermal-effusion flux, and electrons emitted by the surface enter the plasma at the thermionic flux Γ_{em}:

$$\boldsymbol{\Gamma}_e\cdot\hat{\mathbf{n}}=\tfrac{1}{4}n_e\bar{c}_e-\Gamma_{\mathrm{em}},\qquad \bar{c}_e=\sqrt{\frac{8k_BT_e}{\pi m_e}}. \tag{16}$$

The emitted flux is given by the Richardson–Dushman law with Schottky lowering of the work function W by the field at the surface,

$$\Gamma_{\mathrm{em}}=\frac{A_RT_{\mathrm{wall}}^2}{e}\exp\!\left(-\frac{W-\Delta W}{k_BT_{\mathrm{wall}}}\right),\qquad \Delta W=\sqrt{\frac{e^3E_w}{4\pi\varepsilon_0}},\qquad E_w=\max(\mathbf{E}\cdot\hat{\mathbf{n}},0), \tag{17}$$

where A_R is the Richardson constant. The lowering acts only when the field points into the electrode, which is the orientation that accelerates emitted electrons away from it. For the anode, $\Gamma_{\mathrm{em}}=0$. Each absorbed electron removes a mean energy $2k_BT_e$ from the plasma, and each emitted electron brings in $2k_BT_{\mathrm{wall}}$, so the boundary condition for the electron energy equation (11) is

$$\mathbf{h}_e\cdot\hat{\mathbf{n}}=2k_BT_e\,\tfrac{1}{4}n_e\bar{c}_e-2k_BT_{\mathrm{wall}}\,\Gamma_{\mathrm{em}}. \tag{18}$$

The ion wall flux depends on the direction of ion transport at the wall. Let $\boldsymbol{\Gamma}_i^d$ denote the ion flux (9) without its density-diffusion term, $\boldsymbol{\Gamma}_i^d=\boldsymbol{\Gamma}_i+[k_BT_h/(M\bar{\nu}_i^M)]\nabla n_i$, which contains the field drift and the thermal, thermal-diffusion, mixture-force, and drag contributions. Ions are absorbed at this flux when it is directed into the wall, and at their thermal-effusion flux otherwise,

$$\boldsymbol{\Gamma}_i\cdot\hat{\mathbf{n}}=\begin{cases}\boldsymbol{\Gamma}_i^d\cdot\hat{\mathbf{n}}, & \boldsymbol{\Gamma}_i^d\cdot\hat{\mathbf{n}}>0,\\ \tfrac{1}{4}n_i\bar{c}_i, & \text{otherwise},\end{cases}\qquad \bar{c}_i=\sqrt{\frac{8k_BT_{\mathrm{wall}}}{\pi M}}. \tag{19}$$

In the first case, which applies where the sheath field drives ions into the electrode, the ion density gradient at the wall vanishes. In the second case, the thermal-effusion flux ensures that ions reaching the wall are still absorbed where transport alone does not carry them into it.

The wall is impermeable to the heavy gas, $\mathbf{u}\cdot\hat{\mathbf{n}}=0$, which is the boundary condition for the heavy-gas momentum equation (15); the pressure on the wall face is taken from the adjacent plasma. Equation (**??**) then gives

$$\boldsymbol{\Gamma}_n\cdot\hat{\mathbf{n}}=-\boldsymbol{\Gamma}_i\cdot\hat{\mathbf{n}}\,: \tag{20}$$

every ion neutralized at the surface re-enters the plasma as an atom. Secondary electron emission by ion impact is not included.

The heavy-particle temperature at the surface equals the electrode temperature, $T_h(0) = T_{\text{wall}}$. For the anode, T_{wall} is prescribed. For the cathode, it is obtained at every time step by balancing the heat flux delivered by the plasma against steady conduction through the cathode body,

$$q_w = \frac{k_c}{L_c}\left(T_{\text{wall}} - T_0\right), \tag{21}$$

where k_c is the thermal conductivity of the cathode, L_c its length, and T_0 the temperature of its cooled base. The heat flux into the surface is

$$q_w = \left(2k_B T_e + W\right)\tfrac{1}{4} n_e \bar{c}_e - \left(2k_B T_{\text{wall}} + W\right)\Gamma_{\text{em}} + \left(\varepsilon_{\text{ion}} - W\right)\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}} - \kappa_h \nabla T_h \cdot \hat{\mathbf{n}}. \tag{22}$$

The first term is the energy delivered by absorbed electrons, their kinetic energy plus the work function released on entering the metal. The second is emission cooling: each emitted electron removes the work function and its kinetic energy. The third is the energy released when an ion recombines at the surface, ε_{ion}, less the work function spent to extract the neutralizing electron. The last is heat conduction from the heavy gas. Emission cooling rises steeply with T_{wall}, while the right-hand side of equation (21) rises linearly, so the balance has a single root, which is found by bisection.

Bulk-plasma edge, $x = L$. The electron and ion densities and both temperatures are set to Dirichlet values from the adjacent LTE bulk plasma: $n_e = n_e^{\text{bulk}}$, $n_i = n_i^{\text{bulk}}$, and $T_e = T_h = T_{\text{bulk}}$; the last of these is the boundary condition for both energy equations. The neutral density follows from Dalton's law at the background pressure p_0,

$$n_n = \frac{p_0 - n_e k_B T_e}{k_B T_h} - n_i. \tag{23}$$

This edge is open to the heavy gas: the heavy-gas momentum equation (15) sees the bulk state, and hence the pressure p_0, outside the boundary, and the heavy-gas velocity is given a zero normal gradient. The arc current is imposed through the electric field in Poisson's equation (10): the total current density at the boundary, conduction plus displacement, is held equal to the applied current density J_{app},

$$e\left(\boldsymbol{\Gamma}_i - \boldsymbol{\Gamma}_e\right) \cdot \hat{\mathbf{e}}_x + \varepsilon_0 \frac{\partial E_x}{\partial t} = J_{\text{app}}, \tag{24}$$

which is integrated in time as an evolution equation for the boundary field. The initial field enters only as the initial condition of this equation.

3 Appendix A. Transport Coefficients

This appendix gives the transport and exchange coefficients that close the governing equations of Section 2.1: the electron momentum-transfer and energy-exchange collision frequencies, thermal-diffusion coefficients, and thermal conductivity; the ion momentum-transfer frequencies; and the heavy-particle heat capacity and thermal conductivity. Every coefficient is a closed-form or tabulated function of the local state.

3.1 Electron transport: general framework

The electron current density $\mathbf{J}_e = -e\boldsymbol{\Gamma}_e$ and heat flux $\mathbf{q}_e$ obey a linear-response (Onsager) relation with two driving forces. These are the generalized field $\boldsymbol{\mathcal{E}} = \mathbf{E} + \nabla p_e/(en_e)$ and the temperature gradient:

$$\mathbf{J}_e = \sigma\,\boldsymbol{\mathcal{E}} + \varphi\,\nabla T_e, \tag{25}$$

$$\mathbf{q}_e = -\frac{5}{2}\frac{k_B T_e}{e}\mathbf{J}_e - \lambda'\nabla T_e - T_e\varphi\,\boldsymbol{\mathcal{E}} = -\lambda\nabla T_e + \psi\,\mathbf{J}_e, \tag{26}$$

with

$$\lambda = \lambda' - T_e\frac{\varphi^2}{\sigma}, \qquad \psi = -\frac{5}{2}\frac{k_B T_e}{e} - T_e\frac{\varphi}{\sigma}. \tag{27}$$

Here σ is the electrical conductivity, φ the thermal-diffusion coefficient, λ' the thermal conductivity at fixed $\boldsymbol{\mathcal{E}}$, and λ the ordinary (zero-current) thermal conductivity. Comparing equations (25)–(27) with the electron momentum balance (2) and the energy flux (12) identifies

$$\bar{\nu}_e^M = \frac{n_e e^2}{m_e \sigma}, \qquad C_e = \frac{e}{k_B}\frac{\varphi}{\sigma}, \qquad \kappa_e = \lambda, \qquad \frac{5}{2} + C_e = -\frac{e\psi}{k_B T_e}. \tag{28}$$

The conductivity entering the energy equation is the zero-current value λ, not λ'. A temperature gradient alone drives a current $\varphi\nabla T_e$, and holding $\mathbf{J}_e = 0$ requires a field $\boldsymbol{\mathcal{E}} = -(\varphi/\sigma)\nabla T_e$. The heat flux carried by this field reduces the apparent conduction from λ' to λ. Because equation (11) accounts separately for Joule heating and the enthalpy flux, its conductivity must be the zero-current one. The coefficients σ, φ, and λ are evaluated in the partially and fully ionized limits and then bridged.

3.2 Partially ionized limit: electron–neutral transport

When electron–neutral collisions dominate, the electrons form a Lorentz gas. The isotropic part of the distribution is a Maxwellian $f_M(T_e)$, and only the first-order anisotropic correction responds to the field and the temperature gradient. With C the electron speed, $x = m_e C^2/(2k_B T_e)$, and $\nu_{eH}(C) = n_n C Q^{(1)}(\varepsilon)$ the energy-dependent electron–neutral momentum-transfer frequency built from the momentum-transfer cross section $Q^{(1)}(\varepsilon)$, a two-term solution of the electron Boltzmann equation gives the anisotropic correction

$$f^1 = \frac{e}{m_e \nu_{eH}} \frac{\partial f_M}{\partial C} \boldsymbol{\mathcal{E}} - \frac{C}{\nu_{eH}} f_M \left[\nabla \ln p_e + \left(x - \tfrac{5}{2} \right) \nabla \ln T_e \right]. \tag{29}$$

Two different kinds of velocity average appear below, and they are given different notation. The first is the plain average over the Maxwellian, normalized to unity,

$$\langle g \rangle_0 = \int g\, f_M\, d^3C = 4\pi \int_0^\infty g\, C^2 f_M\, dC. \tag{30}$$

The second is the *transport average*. Every current and conduction heat-flux moment of equation (29) carries, besides the Maxwellian, an extra factor C^2/ν_{eH}, so the transport coefficients are expressed through averages under the weight $C^4 f_M/\nu_{eH}$:

$$\langle g \rangle_\nu = \frac{\langle g\, C^2/\nu_{eH} \rangle_0}{\langle C^2/\nu_{eH} \rangle_0} = \frac{\int_0^\infty g\, C^4 f_M \nu_{eH}^{-1}\, dC}{\int_0^\infty C^4 f_M \nu_{eH}^{-1}\, dC}. \tag{31}$$

In energy, this weight is $\varepsilon\, e^{-\varepsilon/k_B T_e}/Q^{(1)}(\varepsilon)$. The transport average is used only for the dimensionless moments $\langle x \rangle_\nu$ and $\langle x^2 \rangle_\nu$ in equation (34) below. It is *not* the average that defines the collision frequencies.

The collision frequencies in the model are Maxwellian averages $\langle \cdot \rangle_0$ of $\nu_{eH}(C)$, and two different ones are needed. Both use the same weight, $C^2 f_M$, that is, the Maxwellian weighted by the electron kinetic energy. They differ in whether ν_{eH} or $1/\nu_{eH}$ is averaged. The *momentum-transfer (mobility) average*, denoted by the superscript M, is the harmonic mean of ν_{eH} under this weight. It reproduces the mobility and is the frequency used in the force balances (2)–(4):

$$\bar{\nu}_{en}^M = \frac{\langle C^2 \rangle_0}{\langle C^2/\nu_{eH} \rangle_0} = n_n \frac{3k_B T_e/m_e}{4\pi \int_0^\infty C^3 f_M \left[Q^{(1)} \right]^{-1} dC} = n_n \sqrt{\frac{2}{m_e}}\, \frac{3\sqrt{\pi}}{4}\, \frac{(k_B T_e)^{5/2}}{\int_0^\infty \varepsilon\, e^{-\varepsilon/k_B T_e} \left[Q^{(1)}(\varepsilon) \right]^{-1} d\varepsilon}. \tag{32}$$

The second form follows from $\nu_{eH} = n_n C Q^{(1)}$ and $\langle C^2 \rangle_0 = 3k_B T_e/m_e$. The third follows from the change of variable $\varepsilon = \frac{1}{2} m_e C^2$, with $f_M = (m_e/2\pi k_B T_e)^{3/2} e^{-\varepsilon/k_B T_e}$. For a constant cross section, it reduces to $\bar{\nu}_{en}^M = (9\pi/32)\, \frac{4}{3} n_n \bar{c}_e Q^{(1)} \approx 0.88 \times \frac{4}{3} n_n \bar{c}_e Q^{(1)}$. Equation (32) is identical to defining $\bar{\nu}_{en}^M$ through the Lorentz conductivity of [36] (their equation VIII 2.14a), $\sigma_{en} = n_e e^2/(m_e \bar{\nu}_{en}^M)$. This is how it is evaluated in practice. The *energy-exchange average*, denoted by the superscript E, is the arithmetic mean of ν_{eH} under the same weight:

$$\bar{\nu}_{en}^E = \frac{\langle C^2 \nu_{eH} \rangle_0}{\langle C^2 \rangle_0} = \frac{4}{3} n_n \bar{c}_e\, \overline{Q}_{en}^{(1,1)}(T_e), \qquad \bar{c}_e = \sqrt{\frac{8k_B T_e}{\pi m_e}}. \tag{33}$$

Because a harmonic mean never exceeds the arithmetic mean over the same weight, $\bar{\nu}_{en}^M \le \bar{\nu}_{en}^E$, with equality only for an energy-independent ν_{eH}. The denominator of equation (32) is also the normalizing integral of the transport average (31). With $\langle C^2 \rangle_0 = 3k_B T_e/m_e$, the Lorentz-gas coefficients are

$$\sigma_{en} = \frac{n_e e^2}{m_e \bar{\nu}_{en}^M}, \qquad \left(\frac{\varphi}{\sigma} \right)_{en} = \frac{k_B}{e} \left(\langle x \rangle_\nu - \frac{5}{2} \right), \qquad \lambda_{en} = \frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{en}^M} \left[\langle x^2 \rangle_\nu - \langle x \rangle_\nu^2 \right]. \tag{34}$$

The electron–neutral thermal-diffusion coefficient $(e/k_B)(\varphi/\sigma)_{en} = \langle x \rangle_\nu - \frac{5}{2}$ vanishes for an energy-independent collision frequency. Its sign and magnitude therefore reflect the energy dependence of $Q^{(1)}$, which for argon is dominated by the Ramsauer minimum.

The energy-exchange average (33) is the one that gives the elastic energy-loss rate of a Maxwellian electron population, $\frac{2m_e}{M}\langle\varepsilon\,\nu_{eH}\rangle_0(1-T_h/T_e)=\frac{3}{2}k_B(T_e-T_h)\frac{2m_e}{M}\bar{\nu}_{en}^E$. For argon between 0.5 and 2 eV, $\bar{\nu}_{en}^E$ exceeds $\bar{\nu}_{en}^M$ by a factor of 4–4.5, because energy exchange is dominated by fast electrons above the Ramsauer minimum, while the mobility is dominated by slow electrons trapped inside it. Using $\bar{\nu}_{en}^E$ in the force balances would therefore overestimate the electron–neutral friction by this factor, and using $\bar{\nu}_{en}^M$ in the energy equations would underestimate the elastic loss by the same factor. Equations (31)–(33) are evaluated on the argon momentum-transfer cross section [37] and tabulated as functions of T_e, in reduced form per unit neutral density.

3.3 Fully ionized limit and the Coulomb logarithm

When electron–ion collisions dominate, the Spitzer–Härm solution for a singly charged ion gives [36]

$$\sigma_{ei}=1.975\,\frac{n_e e^2}{m_e\bar{\nu}_{ei}^E},\qquad \lambda_{ei}=3.203\,\frac{n_e k_B^2 T_e}{m_e\bar{\nu}_{ei}^E},\qquad \varphi_{ei}=1.389\,\frac{n_e e k_B}{m_e\bar{\nu}_{ei}^E},\qquad \psi_{ei}=-3.203\,\frac{k_B T_e}{e}. \tag{35}$$

The collision frequency cancels in $(e/k_B)(\varphi/\sigma)_{ei}=1.389/1.975=0.7033$, and the consistency condition of equation (28), $-e\psi/(k_BT_e)=3.203=\frac{5}{2}+0.703$, is satisfied. Here $\bar{\nu}_{ei}^E$ is the energy-exchange (C^2-weighted) average of the electron–ion momentum-transfer frequency,

$$\bar{\nu}_{ei}^E=n_i\,\frac{4\sqrt{2\pi}}{3}\left(\frac{m_e}{k_BT_e}\right)^{3/2}\left(\frac{e^2}{4\pi\varepsilon_0 m_e}\right)^2\ln\Lambda=2.906\times10^{-12}\,\frac{n_i[\mathrm{m}^{-3}]\,\ln\Lambda}{T_e[\mathrm{eV}]^{3/2}}\ \mathrm{s}^{-1}, \tag{36}$$

with the Coulomb logarithm

$$\ln\Lambda=\max\left(2,\ \ln\frac{\lambda_{De}}{b_{90}}\right),\qquad \lambda_{De}=\sqrt{\frac{\varepsilon_0 T_e[\mathrm{eV}]}{n_e e}},\qquad b_{90}=\frac{e}{12\pi\varepsilon_0 T_e[\mathrm{eV}]}. \tag{37}$$

Here b_{90} is the classical 90°-deflection impact parameter. The logarithm is floored at 2 to keep $\bar{\nu}_{ei}^E$ finite when the plasma is not weakly coupled. The rate (36) is set by the ion density, which is the collision partner, whereas the Debye length uses the electron density. The two coincide only under quasi-neutrality, which does not hold inside the sheath.

3.4 Bridging the two limits

At intermediate ionization, the electron–neutral and electron–ion contributions are combined by additive (resistance-type) mixture rules. These rules are exact in both limits and are weighted by the ratio of the energy-exchange averages of the two collision frequencies,

$$P=\frac{\bar{\nu}_{en}^E}{\bar{\nu}_{ei}^E}. \tag{38}$$

Momentum-transfer frequency. The electron–ion momentum-transfer average is $\bar{\nu}_{ei}^M=0.5\,\bar{\nu}_{ei}^E$, where the factor $0.5\approx1/1.975$ is the strongly ionized conductivity weight of equation (35). Adding the electron–neutral and electron–ion resistivities is the same as adding the two momentum-transfer frequencies,

$$\bar{\nu}_e^M=\bar{\nu}_{en}^M+\bar{\nu}_{ei}^M,\qquad \sigma=\frac{n_e e^2}{m_e\bar{\nu}_e^M}, \tag{39}$$

with $\bar{\nu}_{en}^M$ from equation (32). This is the total frequency in the electron flux (8). Relative to the electron–neutral value alone, the electron–ion (Coulomb) contribution reduces the conductivity by the factor $\bar{\nu}_{en}^M/\bar{\nu}_e^M$.

Thermal-diffusion coefficients. The two values of φ/σ are combined as a weighted mean in $1/P$:

$$\left(\frac{\varphi}{\sigma}\right)_e=\frac{(\varphi/\sigma)_{ei}/P+(\varphi/\sigma)_{en}}{1+1/P},\qquad \left(\frac{\varphi}{\sigma}\right)_{ei}=\frac{1.389}{1.975}\,\frac{k_B}{e}, \tag{40}$$

so that

$$C_e=\frac{e}{k_B}\left(\frac{\varphi}{\sigma}\right)_e,\qquad C_i=\frac{e}{k_B}\,\frac{(\varphi/\sigma)_{ei}/P}{1+1/P},\qquad C_n=C_e-C_i. \tag{41}$$

Only electron–ion collisions couple momentum directly between electrons and ions. The ion reaction C_i is therefore the electron–ion (Coulomb) part of the thermal-diffusion force alone. The

electron–neutral part acts on the atoms, giving $C_n = C_e - C_i$, and the sum rule $C_i + C_n = C_e$ used in Section 2.1 holds identically.

Thermal conductivity. The zero-current conductivities combine as parallel thermal resistances,

$$\frac{1}{\lambda_e} = \frac{1}{\lambda_{en}} + \frac{1}{\lambda_{ei}}, \qquad \kappa_e = \lambda_e \equiv K\,\frac{n_e k_B^2 T_e}{m_e \bar{\nu}_e^M}, \tag{42}$$

with $\bar{\nu}_e^M$ the total momentum-transfer frequency of equation (39). Both λ_{en} and λ_{ei} scale linearly with n_e, so the dimensionless factor K is independent of the electron density. It tends to $\langle x^2\rangle_\nu - \langle x\rangle_\nu^2$ in the electron–neutral limit and to $3.203\,\bar{\nu}_{ei}^M/\bar{\nu}_{ei}^E = 3.203/2 \approx 1.60$ in the fully ionized limit.

Elastic exchange frequency. The total elastic energy-exchange frequency in equations (11) and (13) is the sum of the two energy-exchange averages, not of the momentum-transfer averages $\bar{\nu}^M$:

$$\bar{\nu}_{eh}^E = \bar{\nu}_{en}^E + \bar{\nu}_{ei}^E. \tag{43}$$

Both terms are, by construction, the elastic energy-loss rate of a near-Maxwellian electron population. The energy lost by the electrons in equation (11) is therefore exactly the energy gained by the heavy gas in equation (13).

3.5 Ion collision frequencies

The ion–neutral transport is governed by resonant charge exchange. The momentum-transfer cross section is taken from the fit [32]

$$\overline{Q}_{in} = 2\left[7\times10^{-10} - 0.6\times10^{-10}\ln\left(\frac{2k_BT_h}{e}\right)\right]^2 \ \mathrm{m}^2, \tag{44}$$

with $2k_BT_h/e$ in eV. The corresponding low-field mobility is

$$\mu_i^0 = \frac{1.104\times10^5}{\overline{Q}_{in}\sqrt{T_h}\,n_n} \ \mathrm{m^2\,V^{-1}\,s^{-1}} \qquad (\overline{Q}_{in}\text{ in m}^2,\ T_h\text{ in K},\ n_n\text{ in m}^{-3}). \tag{45}$$

The high reduced fields near the electrode heat the ions. This is accounted for by a closed-form two-temperature Wannier formulation [38], in which the drift speed v_d and the effective ion temperature T_{eff} are determined self-consistently:

$$v_d^2 = \frac{\sqrt{1+4\mathcal{A}\mathcal{B}}-1}{2\mathcal{B}}, \qquad \mathcal{A} = \left(\mu_i^0|\mathbf{E}|\right)^2, \qquad \mathcal{B} = \frac{M}{3k_BT_h}, \tag{46}$$

$$T_{\mathrm{eff}} = T_h + \frac{Mv_d^2}{3k_B}, \qquad \mu_{in} = \mu_i^0\sqrt{\frac{T_h}{T_{\mathrm{eff}}}}. \tag{47}$$

Equation (46) is the positive root of the quadratic in v_d^2 obtained by substituting $\mu_{in}(T_{\mathrm{eff}})$ into $v_d = \mu_{in}|\mathbf{E}|$. In the low-field limit, $v_d \to \mu_i^0|\mathbf{E}|$, $T_{\mathrm{eff}} \to T_h$, and $\mu_{in} \to \mu_i^0$. The charge-exchange data are available in this mobility form. They enter the model only through the ion–neutral momentum-transfer frequency

$$\bar{\nu}_{in}^M = \frac{e}{M\mu_{in}}. \tag{48}$$

The ion–electron frequency follows from momentum conservation, equation (5),

$$\bar{\nu}_{ie}^M = \frac{m_e n_e}{M n_i}\,\bar{\nu}_{ei}^M, \tag{49}$$

with $\bar{\nu}_{ei}^M = 0.5\,\bar{\nu}_{ei}^E$ as in equation (39). Equations (48) and (49) give the total ion momentum-transfer frequency $\bar{\nu}_i^M = \bar{\nu}_{in}^M + \bar{\nu}_{ie}^M$, which sets the ion mobility and diffusivity in equation (9). In the weakly ionized bulk, $\bar{\nu}_{ie}^M \ll \bar{\nu}_{in}^M$, so $\bar{\nu}_i^M \to \bar{\nu}_{in}^M$. As $n_n \to 0$ (near-total ionization), $\bar{\nu}_{in}^M \propto n_n$ vanishes and $\bar{\nu}_i^M$ is bounded below by the Coulomb-collision value $\bar{\nu}_{ie}^M$, which keeps the ion mobility finite. The ion–neutral frictional heating (14) uses the charge-exchange frequency $\bar{\nu}_{in}^M$ of equation (48) alone, not $\bar{\nu}_i^M$, which also contains the electron-friction contribution.

3.6 Heavy-particle heat capacity and thermal conductivity

The heat capacity per heavy particle in equation (13) contains only the translational degrees of freedom, $c_v = \frac{3}{2}k_B$. The gas is compressible, and the work of expansion appears explicitly as $-n_h k_B T_h \nabla\cdot\mathbf{u}$. The electron energy is carried separately, and the ionization energy enters as a source in the electron energy equation. The heavy-particle thermal conductivity is the sum of an atom and an ion channel, each given by a first-approximation Chapman–Enskog expression:

$$\kappa_h = \kappa_n + \kappa_i, \tag{50}$$

$$\kappa_n = \frac{75k_B}{64}\sqrt{\frac{\pi k_B T_h}{M}}\,\frac{n_n}{n_n\overline{Q}_{nn} + n_i\overline{Q}_{in}}, \qquad \kappa_i = \frac{75k_B}{64}\sqrt{\frac{\pi k_B T_h}{M}}\,\frac{n_i}{n_n\overline{Q}_{in} + n_i\overline{Q}_{ii}}, \tag{51}$$

with the viscosity-type $(2,2)$ cross sections

$$\overline{Q}_{nn}^{(2,2)} = 1.12\times10^{-18}\,T_h^{-0.2}, \qquad \overline{Q}_{in}^{(2,2)} = 3.6\times10^{-18}\,T_h^{-0.3}, \qquad \overline{Q}_{ii}^{(2,2)} = \frac{e^4\ln\Lambda}{36\pi\varepsilon_0^2(k_BT_h)^2}, \tag{52}$$

in m^2 with T_h in K. Here $\ln\Lambda$ is the Coulomb logarithm of equation (37). The expression is regular in both limits. As $n_n \to 0$, κ_i tends to the pure ion–ion Coulomb value, and as $n_i \to 0$, κ_n tends to the neutral–neutral value.

3.7 Reaction rate coefficients

The ionization and three-body recombination rate coefficients $k_{\mathrm{ion}}(T_e)$ and $k_{\mathrm{rec}}(T_e)$ in equation (**??**) are effective rate coefficients from Hoffert and Lien [33]. They are tabulated in T_e and interpolated log-linearly. The energy charged to the electrons per net ionization event is $\varepsilon_{\mathrm{ion}} = 15.76$ eV.

3.8 Quasi-Maxwellian electrons near the electrode

Even where the heavy species are strongly out of equilibrium, the electron population can remain closely Maxwellian provided electron–electron and electron–ion collisions are frequent enough to redistribute directed (drift) energy into the isotropic thermal spread faster than electron–heavy collisions or the imposed field can distort it. Following Dinulescu and Pfender's analysis of the anode boundary layer [19], two conditions must hold. First, electron–electron/ion thermalization must dominate electron–neutral momentum transfer, weighted by the mass ratio that makes each electron–neutral collision inefficient at removing energy:

$$\sum_h \frac{m_e}{m_h}\,\frac{\bar\nu_{eh}}{\bar\nu_{ee}} \ll 1. \tag{53}$$

Second, the ordered drift energy imparted by the field must remain small compared with the electron thermal energy, weighted by the same mass ratio:

$$\left(\frac{U_e}{\bar c_e}\right)^2 \ll \frac{m_e}{m_h}. \tag{54}$$

For a representative near-anode condition, $T_e \sim 1$ eV, $n_e \sim 5\times10^{22}\,\mathrm{m}^{-3}$, neutral density $n_n \sim 6\times10^{23}\,\mathrm{m}^{-3}$, electron–neutral momentum-transfer cross section $\bar Q_{en} = 2\times10^{-20}\,\mathrm{m}^2$, and mean thermal speed $\bar c_e = 7\times10^5$ m/s: the electron–ion/electron–electron frequency from the NRL formulary (with $\ln\Lambda \approx 3$) is $\nu_{ee} \sim \nu_{ei} \sim 5\times10^{11}\,\mathrm{s}^{-1}$, while the electron–neutral frequency is $\nu_{en} = \bar c_e n_n \bar Q_{en} \approx 8\times10^9\,\mathrm{s}^{-1}$, i.e. $\nu_{en}/\nu_{ee} \sim 0.016$; with $m_e/M_{\mathrm{Ar}} \approx 1.4\times10^{-5}$, condition (53) gives $\sim 2\times10^{-7} \ll 1$. For condition (54), the electrical conductivity of the layer, $\sigma \approx 5000$ S/m, corresponds to an electron momentum-transfer frequency $\bar\nu_e^M = e^2 n_e/(m_e\sigma) \approx 3\times10^{11}\,\mathrm{s}^{-1}$; with a representative current density $J \sim 10^7\,\mathrm{A/m}^2$, the electron drift speed is $U_e = J/(en_e) \approx 1.3\times10^3$ m/s, so $(U_e/\bar c_e)^2 \approx 3\times10^{-6}$, again well below $m_e/M_{\mathrm{Ar}} \approx 1.4\times10^{-5}$. Both conditions are comfortably satisfied, so the electron population is treated as quasi-Maxwellian throughout the near-electrode layer, and a single electron temperature T_e (rather than a fully kinetic electron energy distribution) is retained in equation (11) there as well as in the bulk.

3.9 Length scales and the thickness of the near-electrode layer

The same parameters fix a hierarchy of length scales that separates the space-charge sheath from the broader near-electrode non-equilibrium region that must be resolved around it. The electron Debye length,

$$\lambda_{De} = \sqrt{\frac{\varepsilon_0 T_e}{n_e e}} \approx 3\times10^{-8}\ \mathrm{m}, \tag{55}$$

sets the scale of the space-charge sheath itself. The electron mean free path against the combined electron–neutral and electron–electron/ion collision frequency used above,

$$\lambda_e = \frac{\bar{c}_e}{\nu_{en} + \nu_{ee}} \approx 1 \times 10^{-6}\ \mathrm{m}, \tag{56}$$

is ~ 30 times λ_{De}, confirming that the sheath is thin compared with a mean free path and can support the non-collisional electron kinetics discussed in Section 3.10. The electron–heavy energy-relaxation length, over which an electron transfers its excess thermal energy to the heavy gas through the inefficient mass-ratio-weighted elastic channel in equation (11), is larger still,

$$\lambda_T = \frac{\lambda_e}{\sqrt{2m_e/M}} \approx 2 \times 10^{-4}\ \mathrm{m}. \tag{57}$$

Finally, the diffusion length that controls how far the electron/ion density profile itself can depart from its local equilibrium value under a given field, used directly by Lowke and Tanaka's mesh-sizing rule [16], is $\Lambda = (kT_e/e)/E$; with $T_e \approx 1$ V and a field $E = J/\sigma \approx (10^7\ \mathrm{A/m^2})/(5000\,\mathrm{S/m}) = 2000$ V/m consistent with the parameters above,

$$\Lambda \approx 5 \times 10^{-4}\ \mathrm{m}. \tag{58}$$

The four scales satisfy $\lambda_{De} \ll \lambda_e \ll \Lambda \sim \lambda_T$, the same ordering (Debye length $\ll$ mean free path $\ll$ boundary-layer thickness) reported directly in Table I of [19] for the anode boundary layer of a comparable arc, $\lambda_{De} \sim (2\text{–}3) \times 10^{-8}$ m, $\lambda_e \sim (1\text{–}2) \times 10^{-6}$ m, boundary-layer thickness $L \sim 0.2$–0.6 mm. On this basis we take a near-electrode layer of thickness $\delta \approx 0.2$ mm — comparable to both Λ and λ_T, and consistent with the range of L found in [19] — as the region over which the unified, non-equilibrium model of Section 4 must be applied; outside it, the plasma is close enough to collision-dominated equilibrium for the LTE bulk treatment of Section to hold. Note that this ∼0.2 mm layer is not itself the space-charge sheath: it is the outer, thermal/diffusion-dominated non-equilibrium region (corresponding to layers 2–3 of the four-sublayer structure found by [24]) inside which a much thinner space-charge sheath, of order λ_{De} to a few times λ_{De}, is nested.

3.10 Is the sheath collisionless?

A collisionless sheath is the standard simplifying assumption used to close the NLTE-sheath and two-temperature ionization-layer-sheath approaches of Section 1.1, and it is tempting to expect it to hold automatically at atmospheric pressure, where collision rates are high. Almeida, Benilov and Naidis show this expectation is wrong: computing the number of ion collisions actually incurred while crossing the sheath gives values from order unity to 10^4, not $\ll 1$, even for atmospheric-pressure argon [24]. The reason is that the sheath thickness itself scales with the *local* Debye length, $\lambda_{De} \propto n_e^{-1/2}$, which is set by the arc's own high electron density and shrinks faster, as the arc becomes hotter and more strongly ionized, than the neutral mean free paths shrink with pressure (the Ramsauer minimum in the argon electron–neutral cross section keeps λ_e from falling as quickly as λ_{De}). A collisionless sheath at high pressure is therefore not evidence of a crude or incomplete model; it is close to the generic outcome for a dense atmospheric-pressure arc, and the question that actually needs answering species by species and quantity by quantity is *how* collisionless, and for what.

We make this quantitative with four Knudsen numbers, each the ratio of the relevant mean free path to the resolved sheath width s: $\mathrm{Kn}_e = \lambda_e/s$ (do electrons collide crossing the sheath at all); $\mathrm{Kn}_\varepsilon = \lambda_T/s$ (electron–heavy energy exchange: does a local electron energy balance exist inside the sheath); $\mathrm{Kn}_{ee} = \lambda_{ee}/s$ (electron–electron thermalization: does a well-defined T_e, i.e. a locally Maxwellian electron energy distribution, exist inside the sheath at all – a distinct and much more restrictive question than Kn_ε, since λ_{ee} and λ_T differ by roughly 10^4); and $\mathrm{Kn}_i = \lambda_i/s$ for the ion charge-exchange mean free path. For the converged, resolved-sheath operating point of the PLASMEX test case discussed in Section 4 ($s = 1.65 \times 10^{-7}$ m $= 5.6\,\lambda_{De}$, $T_e \approx 0.98$ eV at the wall), these evaluate to $\mathrm{Kn}_e \approx 63$, $\mathrm{Kn}_\varepsilon \approx 1.2 \times 10^4$, $\mathrm{Kn}_{ee} \approx 5.3$, and $\mathrm{Kn}_i \approx 0.8$–$1.7$: the sheath is collisionless for electron momentum and electron–heavy energy exchange by a wide margin, only marginally collisionless for electron thermalization, and carries of order one ion collision, consistent with the lower end of the 10^2–10^4 range quoted by [24] for a somewhat different case. Despite $\mathrm{Kn}_{ee} \sim 5$ being only marginal, an electron population entering a purely retarding, elastic potential barrier is mapped onto itself by Liouville's theorem: reflection by an energy-conserving barrier preserves a Maxwellian at unchanged T_e while depleting the density by the Boltzmann factor,

giving a testable, parameter-free reference prediction, $T_e(x) = T_{e,\mathrm{edge}} = \text{const}$ across the sheath, against which the local closure used in equation (11) can be checked directly (Section 4). No such collisionless reference state exists for the ions, which do not see a retarding barrier and for which $\mathrm{Kn}_i = O(1)$ rules out both the collisionless limit and full collisional equilibrium; ion behaviour in the sheath is therefore treated as an explicit target for the resolved drift-diffusion solution rather than assumed a priori in either limit.

4 Unified near-electrode model and numerical implementation

4.1 Resolving the sheath directly, rather than matching or homogenizing it

Section 1.1 distinguished three published strategies for coupling a bulk-plasma closure to the electrode: solving one boundary value problem with Poisson's equation over the whole gap [24]; matching a non-equilibrium bulk solution directly to an analytic sheath closure at an interface, valid strictly only for a collision-dominated sheath [23]; and embedding the unresolved sheath as an effective electrical conductivity in the boundary-adjacent finite-volume cells of an otherwise quasi-neutral CFD solution, $\sigma_{\mathrm{eff}} = [(1 - I_g)/\sigma_{\mathrm{child}} + I_g/\sigma]^{-1}$ with σ_{child} a Child-law space-charge-limited conductivity [31]. We use a fourth strategy, implemented in the PLASMEX code: solve the same drift-diffusion– Poisson system, equations (1)–(10), in the bulk plasma and through the sheath alike, and use block-structured adaptive mesh refinement (AMR), built on the AMReX framework, to refine the mesh at the electrode down to a fraction of the local Debye length. No separate sheath closure, matching condition, or homogenized conductivity is introduced; the degree of local non-equilibrium is a property of the converged solution, not a modelling choice made in advance. This is closest in spirit to the unified modelling of [24], generalized from a specialized one-dimensional boundary-value solver to a general-purpose, AMR finite-volume PDE code, and it should be contrasted directly with the homogenized effective-conductivity shortcut of [31], which is the more practical choice for genuinely three-dimensional geometries where literal sub-Debye-length mesh refinement is not affordable.

4.2 Transport closures

Electron mobility and diffusion coefficients are obtained from a pre-tabulated, effectively Maxwellian two-term Boltzmann solution, $\mu_e N(T_e)$ and $D_e N(T_e)$. This is verified, not assumed: across the full tabulated range the ratio $(D_e/\mu_e)/T_e$ deviates from unity by 0.10% rms (worst case 0.50%), which is exactly the Einstein relation for a Maxwellian electron energy distribution, and confirms that at the 0.9–10% ionization fractions relevant here (ν_{ei}/ν_{en} ranging from 12 to 143 over $T_{\mathrm{gas}} = 1000$–$12\,000$ K) electron–electron and electron–ion collisions dominate every inelastic channel strongly enough to Maxwellize the bulk electron population, exactly as required by Section 3.8. The electron–ion Coulomb correction, equation (39), reduces the neutral-collision-only conductivity by one to three orders of magnitude and corrects its qualitative trend (neutral-only conductivity rises with temperature only because it divides by a falling neutral density; the corrected, Spitzer-like conductivity, $\propto T_e^{3/2}/\ln\Lambda$, falls in the $\sim 10^3$ S/m range expected for LTE argon at 1 atm). A Saha-equilibrium cross-check on the arc-core initial condition, $n_e^{\mathrm{Saha}} = 9.909 \times 10^{22}\,\mathrm{m}^{-3}$ against the solved $n_e = 9.860 \times 10^{22}\,\mathrm{m}^{-3}$, agrees to 0.5%, confirming the bulk solution is a genuine LTE state without that equilibrium being imposed. Ion transport uses the closed-form two-temperature Wannier closure of Section 2.1; at the reduced field found at the converged electrode cell of the sheath test case below (5976 Td), it gives $\mu/\mu_0 = 0.16$, a factor-of-six reduction from the low-field mobility, and recovers $\mu \to \mu_0$ in the low-field bulk.

4.3 Electrode boundary conditions

At the electrode face, the electron flux is the kinetic wall flux $\frac{1}{4} n_e \bar{c}_e$ evaluated on the (self-consistently depleted) wall density, less the Richardson–Schottky emission flux (17) in the cathode case, and the ion flux follows equation (19): the non-diffusive part of the ion flux (9) when it is directed into the wall, and the thermal-effusion flux $\frac{1}{4} n_i \bar{c}_i$ at the wall temperature otherwise. Each wall flux, including the electron energy flux (18), is imposed on the implicit diffusion flux as the difference between the prescribed wall flux and the explicit part, so that the total wall flux is exactly the prescribed one; for the ions, in the drift-dominated case, this makes the density gradient at the wall face zero. In the cathode case the wall temperature is re-solved from the heat balance (21) at the start of every time step, from the state at that time. Two simplifications remain: the electron wall flux uses the field-independent $\frac{1}{4} n_e \bar{c}_e$ form rather than a Hagelaar-type field-aware form, and secondary electron emission by ion impact is not included.

4.4 Numerical implementation

The model is implemented in PlasmeX, a C++17 drift-diffusion plasma code built on the AMReX block-structured AMR framework, currently configured for a three-species (e^-, Ar^+, Ar) argon plasma. The electrostatic potential is solved with AMReX's geometric multigrid solver (MLMG) in general, and with Hypre's structured multigrid solver (`HYPRE_StructPFMG`) for the anisotropic two-dimensional axisymmetric mesh used in production runs, since MLMG's composite multigrid cycle was found not to converge on that mesh at any tested aspect ratio, bottom solver, or coarsening depth. Species transport uses a semi-implicit, operator-split scheme: drift fluxes are upwinded explicitly and diffusive fluxes are backward-Euler in an implicit face-coefficient form, $1 - \Delta t\,(e/\varepsilon_0)\mu_e n_e$, that removes the explicit dielectric-relaxation time-step restriction of a fully explicit electron drift treatment and gives roughly a 100× larger stable time step at equal accuracy, verified by matching both schemes at matched physical time to $\sim 10^{-6}$ relative agreement after a long integration from a common checkpoint.

In one spatial dimension, both linear solves (the electrostatic Poisson solve and the implicit species-diffusion solve) reduce to a tridiagonal system on each AMR level, for which a direct solver is both faster and more accurate than the general-purpose iterative multigrid solver: instrumented at 93% of a time step for a 512-cell problem, the iterative solver is replaced by a direct Thomas-algorithm solve whose matrix entries are recovered once, exactly, by probing the discrete operator, giving an 8.9× reduction in total time-step cost (65× and 13× on the potential and species solves individually) with an exact rather than iteratively converged linear-algebra answer. This extends, non-trivially, to the full composite AMR hierarchy: refluxing at a coarse-fine interface exactly cancels the covered coarse cell and replaces it with the fine-level flux, and the coarse-to-fine ghost-cell interpolation used in one dimension is an exact two-term closure over a single coarse cell, ghost $= \frac{2}{3}x_C + \frac{1}{3}x_{f_0}$ at a refinement ratio of 2 — together these guarantee that the composite operator over the leaf cells of a refined 1D hierarchy is itself exactly tridiagonal, so the same direct solve applies unmodified across level boundaries. This direct solver is what makes literal AMR refinement to sub-Debye-length cells in the sheath computationally practical rather than prohibitively expensive; it is restricted to one spatial dimension and a single MPI rank, so the production two-dimensional axisymmetric configuration falls back to Hypre as noted above.

5 Results

6 Conclusion

Acknowledgments

To be completed.

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